20. THE EYE AND CRANIAL DEVELOPMENT

DURATION : 04:35

STUDY SHEET

COURSE SUMMARY

In this video, we explore in detail the complex links between eye development and skull formation. The focus is on embryological processes, such as primitive cephalic flexion and their impact on bone structure, particularly the central role of the sphenoid in eye stability. The concepts of cephalic growth and orbital spasm are discussed, highlighting the importance of maintaining harmony between the eye, nose, and palate formation. In addition, the video illustrates the developmental spirals that influence cranial structure and their repercussions on visual function, thus providing practical tools for students and therapists wishing to deepen their understanding of osteopathy and biodynamic embryology.

THE EYE AND CRANIAL DEVELOPMENT

The development of the **eye** is intimately linked to the formation of the **cranium**. During **primitive cephalic flexion**, the **sub-mesencephalic mesenchyme** is compressed between the anterior and posterior parts of the **cerebral vesicle**. This process leads to water loss and the transformation of the mesenchyme into **cartilaginous tissue**, thus forming the base of the skull.

At the top, we find the **ethmoid**, followed by the **body of the sphenoid**, and then the **basilar part of the occipital bone**. The **wings of the sphenoid** play a crucial role as a fulcrum for the eye. The latter is supported by the sphenoid, which ensures its **stability**. As the cranium grows, the bony tissue of the eye develops in response to the growth of the **brain**, engramming itself around the eye and closing with specific lines of force.

It is essential to understand that the eye and the cranium are interconnected. A slight impact can destabilise the **orbit** and affect visual **reflection**. Thus, it is crucial to restore the stability of the eye, both through the brain and the cranium.

As the nose develops, the eye appears on the side. Simultaneously, the **palate** forms. The closure of the **palatine** coincides with the alignment of the eye in its vertical and horizontal axes, marking an important terminal point in bone development.

Cephalic growth is paramount. It is at this moment that the eyes move closer to the **midline**. A fulcrum forms at the **nasion**. As cortical development progresses, the eye orientates into its final position.

Two spirals are observed: an eye spiral and a bony counter-spiral, visible in the shape of the cranial **sutures**. The orientation of these sutures is essential for studying the three-dimensionality of the **premaxilla**, which has a complex shape, contributing to the orbital and ocular structure.

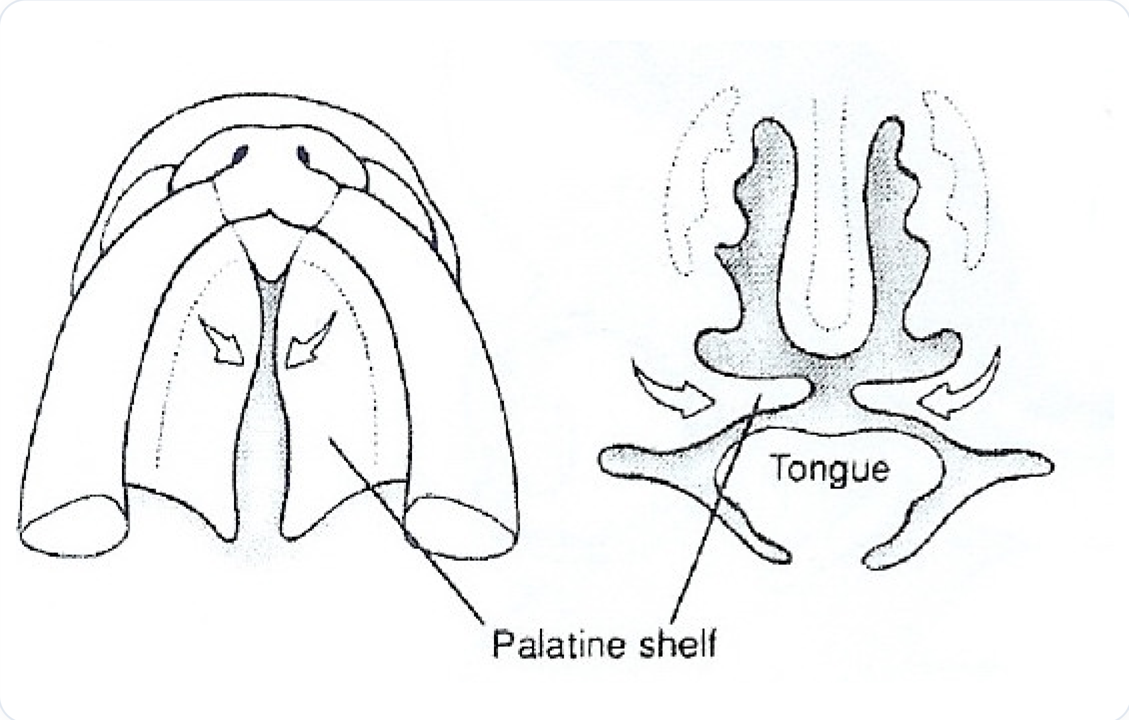

FIG. — Explanatory diagram